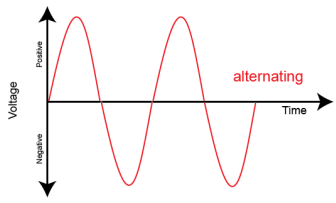
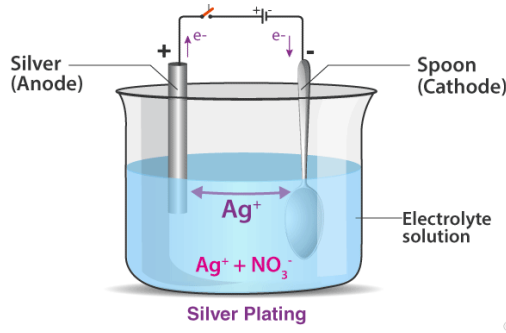


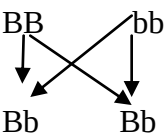
SEE 2082 (2026)
अनिवार्य विज्ञान तथा प्रविधि
उत्तरकुन्जिका

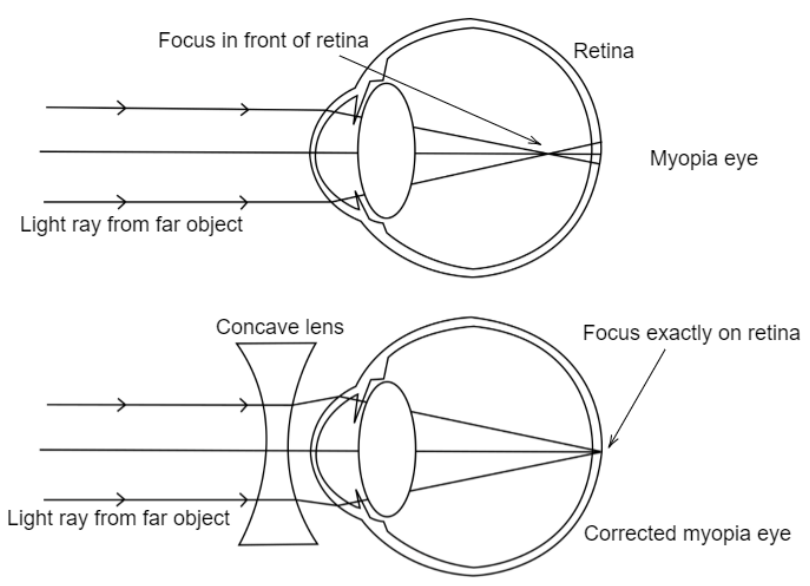
यस कुन्जिकामा भएको बाहेक अन्य सही वैकल्पिक तरिकाबाट समस्या समाधान गरेमा पनि अंक प्रदान गर्नुपर्ने छ ।
 उत्तरकुन्जिकामा प्रत्येक चरणको प्राप्ताङ्क १ प्रदान गर्नुपर्नेछ तथा एकाइ छुट भएमा ०.५ अङ्क मात्र घटाउनुपर्ने छ ।

प्र. नं.	उत्तर	अंक
खण्ड 'क' (Section 'A')		
1a	(i) Because their bodies are externally covered by hard calcareous spines.	1
b	(iv) In a hive the queen bee lives the longest.	1
c	(ii) Asiatic pennywort	1
d	(i) 124.7 N	1
e	(i) The temperature of water in ocean does not vary much between day and night	1
f	(ii) Real, inverted and magnified	1
g	(ii) 35V	1
h	(iii) H_2CO_3	1
i	(iv) $C_n H_{2n}$	1
j	(iii) Calcium Carbide	1
खण्ड 'ख' (Section 'B')		
2.a	To ensure a fair test and determine if the independent variable is the actual cause of the results. By keeping other variables constant, you can be sure that any changes observed are due only to the factor being tested. Or any other correct definition will get 1 mark	1
b	- To pay bills - To share the link of any contents Any one correct answer gets 1 mark	1
c	- It respire through general body surface	1
d	Collects nectar, pollen, water Any one gets 1 mark	1
e	Because it has limited temperature range if exposed at high temperature gives in accurate reading.	1
f	- Figure A is based on Faraday's law of electromagnetic induction - Figure B motr effect	1
g	 - Or any other correct figure	1

2.h			1
2.i	Magnetic separation process		1
खण्ड 'ग' (Section 'C')			
3	Dependent and independent variable Example while writing on paper speed of writing is dependent variable the size of the letter is independent variable or any other correct answer gets 1 mark		1 1
4	- Fish - They live in water - Two chambered	- Frog - They live both on land and water - Three chambered heart	1 1
5	Figure A It is RNA and has single strand It helps in protein synthesis	Figure B It is DNA and has two strands It helps in transference of hereditary character	1
6	The chart denotes the sex determination 44, XX is female, and 44, XY is male		1 1
7	it is related to heart attack preventive measures a. Daily physical exercise b. Avoid smoking and drinking alcohol c. Live stress free life d. Participate recreational activities Any two of them gets 1 mark		1 1
8	- To remove dandruff from the hair - Bone joints - Cracking of the skin - To treat the rough skin - Any two answers get 2 mark		1 1
9	- Law of floatation states that, the floating body displace the equal amount of liquid as of its weight. So, if the mass of the object is more than that of the upthrust applied on it, it sinks - Or any other correct answer get 2 marks		1+1
10	• It shows the motor effect or it moves • The needle does not show any movement		1 1

11	For Or, $10 \times 4200 \times (90 - T) = 20 \times 4200 \times (T - 20)$ Or, 43.33°C	1 1						
12	When the ray of light travels from denser to rarer medium the ray reflects in the same or denser medium i.e. total internal reflection occurs Uses: Wind pipe Optical fibers Any one of the correct use gets 1 mark	1 1						
13	<table><tr><td>Open universe</td><td>Closed universe</td></tr><tr><td>If the average density is less than the critical density</td><td>If the average density is more than the critical density</td></tr><tr><td>Universe is expanding continuously</td><td>The universe is shrink continuously</td></tr></table>	Open universe	Closed universe	If the average density is less than the critical density	If the average density is more than the critical density	Universe is expanding continuously	The universe is shrink continuously	1 1
Open universe	Closed universe							
If the average density is less than the critical density	If the average density is more than the critical density							
Universe is expanding continuously	The universe is shrink continuously							
14	a. Acid base reaction/neutralization b. $\text{NH}_4\text{OH} + \text{HCl} \rightarrow \text{NH}_4\text{Cl} + \text{H}_2\text{O}$	1 1						
15	a. Propan - 1, 2, 3 - triol b. propane	1 1						
16	Herbicide: butachlor, isoproturon, atrazine (any one) Fungicide: Dimethomorph, sectin, mancozeb Carbendazin(any one)	1 1						
खण्ड “घ” Section “D”								
17	Positive - fast communication, - learning is easy - digital library Or Any other correct response Negative - Health problem - Social and psychological problem Or Any other correct response	1 1 1 1						

18	First filial generation  All offsprings are seen black because black character is the dominant character. According to the law of dominance the observation appears. <table border="1" data-bbox="327 414 1300 533"> <tr> <th>Gametes</th><th>B</th><th>b</th></tr> <tr> <th>B</th><td>BB</td><td>Bb</td></tr> <tr> <th>b</th><td>Bb</td><td>bb</td></tr> </table> Phenotypic ratio: 3:1 Genotypic ratio: 1:2:1	Gametes	B	b	B	BB	Bb	b	Bb	bb	1 1 1 1
Gametes	B	b									
B	BB	Bb									
b	Bb	bb									
19	a. Part A b. Part D c. Tricuspid valve d. To distribute oxygenated blood to all body parts	1 1 1 1									
20	a. $v = u + at = 0 + 9.8 \times 4 = 39.2 \text{ m/s}$ or 40 m/s b. $F = G \frac{m_1 \times m_2}{d^2}$ $= 6.67 \times 10^{-11} \frac{4,500 \times 2000}{25 \times 10^{10}}$ $= 24 \times 10^{-16} \text{ N}$	1 1 1 1									

21	a. Short sightedness  b. The ciliary muscle can not be elongated or lens becomes thinner or the focal length of the lens becomes shorter any one	1 1 1
22	a. Y and M b. B size is larger due to which easy to lose electron c. $2\text{Na} + \text{F}_2 \longrightarrow 2\text{NaF}$	1 1 1+1
23	a. Carbondioxide or CO_2 b. Due to heavier than air c. $2\text{Mg} + \text{CO}_2 \longrightarrow 2\text{MgO} + \text{C}$ d. $\text{CO}_2 + \text{NH}_3 \longrightarrow \text{NH}_2\text{-CO-NH}_2 + \text{H}_2\text{O}$	1 1 1 1